

will also be much smaller and x will be smaller.

7 (a) (i) $\langle V^2 \rangle = \frac{(25.0 \times 1.0)}{4.0} = 6.25$
 $V_{rms} = 2.50 \text{ V}$

(ii) $\langle P \rangle = \frac{V_{rms}^2}{R} = \frac{2.50^2}{8.0} = 0.781 \text{ W}$

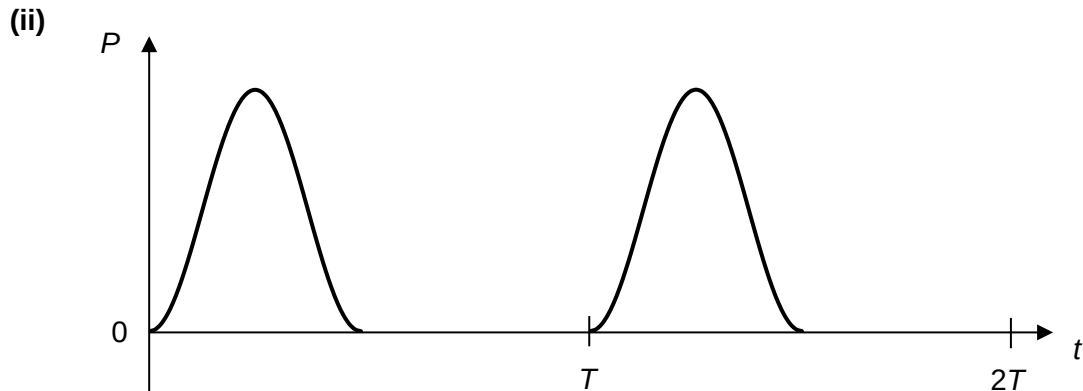
(b) (i) Since average power is the same,
 V_{rms} of sinusoidal function = V_{rms} of original source = 2.50 V
 $V_0 = 2 \times 2.50 = 5.00 \text{ V}$
 Or

For $\frac{1}{2}$ -wave rectified sinusoidal function, $V_{rms} = \frac{V_0}{2}$

$$\langle P \rangle = \frac{V_{rms}^2}{R}$$

$$0.781 = \frac{(V_0 / 2)^2}{8}$$

$$V_0 = 5.00 \text{ V}$$



8

9 (a) (i) While the speed is constant the direction of the velocity is changing